

Brussels, 19 May 2025

COST 040/25

## DECISION

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Subject: Memorandum of Understanding for the implementation of the COST Action “European Network for radiation-detection based Research and Innovation addressing increasing societal CHallenges” (ENRICH) CA24131

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The COST Member Countries will find attached the Memorandum of Understanding for the COST Action European Network for radiation-detection based Research and Innovation addressing increasing societal CHallenges approved by the Committee of Senior Officials through written procedure on 19 May 2025.

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## **MEMORANDUM OF UNDERSTANDING**

For the implementation of a COST Action designated as

**COST Action CA24131**  
**EUROPEAN NETWORK FOR RADIATION-DETECTION BASED RESEARCH AND INNOVATION**  
**ADDRESSING INCREASING SOCIETAL CHALLENGES (ENRICH)**

The COST Members through the present Memorandum of Understanding (MoU) wish to undertake joint activities of mutual interest and declare their common intention to participate in the COST Action, referred to above and described in the Technical Annex of this MoU.

The Action will be carried out in accordance with the set of COST Implementation Rules approved by the Committee of Senior Officials (CSO), or any document amending or replacing them.

The main aim and objective of the Action is to build a European network of researchers and stakeholders working on radiation-detection based experiments and applications, to strengthen connections among different radiation detection physics sub-communities, creating a cross-border, integrated community where multidisciplinary knowledge, information exchange, and discussion of new ideas and challenges are central. This will be achieved through the specific objectives detailed in the Technical Annex.

The present MoU enters into force on the date of the approval of the COST Action by the CSO.

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## OVERVIEW

### Summary

The ENRICH (European Network for radiation-detection based Research and Innovation addressing increasing societal Challenges) COST Action aims to build a **European network of researchers and stakeholders working on radiation-detection based experiments and applications**.

Numerous researchers work with photon detection daily, applying it in astrophysics, nuclear and atomic physics, material science, data analysis, simulations, detectors, electronics, optics, medicine, cultural heritage, food safety, fraud detection and others. However, **the community remains fragmented**, with sub-communities (e.g., nuclear physics, synchrotron users) often working in silos. This **lack of cross-disciplinary collaboration limits knowledge exchange** and the transfer of technologies from large-scale facilities to smaller labs and industries.

ENRICH seeks to unite these diverse communities, allowing **collaboration across sectors and disciplines** to accelerate scientific and technological advances in radiation detection.

Such a collaboration will, first of all, focus on **pressing scientific questions** related to radiation detection in research fields like fundamental physics, nuclear and astrophysics, data analysis, and X- and Gamma-ray spectroscopy or imaging.

In addition, the scientific outcomes of ENRICH's activities will enhance **Europe's industrial competitiveness** by contributing to tackling key societal and technological challenges Europe faces, such as energy autonomy, food security, faster medical diagnostics, and more. These challenges require, indeed, **advances in analytical sciences**, with radiation-based technologies playing a crucial role across various sectors of science and society.

ENRICH will be **grounded on inclusiveness**, and special attention will be paid to young researchers, those with caregiving responsibilities, and participants from ITC countries, through mentoring programs, training in proposal submissions, and special initiatives.

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| <p><b>Areas of Expertise Relevant for the Action</b></p> <ul style="list-style-type: none"> <li>Physical Sciences: Instrumentation - telescopes, detectors and techniques</li> </ul> | <p><b>Keywords</b></p> <ul style="list-style-type: none"> <li>Radiation Detection based applications</li> <li>X-ray and Gamma-ray detectors and electronics</li> <li>Imaging and spectroscopic applications</li> <li>Medical and societal applications</li> <li>Optics, photon sources and MC simulations</li> </ul> |
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### Specific Objectives

To achieve the main objective described in this MoU, the following specific objectives shall be accomplished:

#### Research Coordination

- Build an overtime increasing broad and multidisciplinary community that includes experts,

institutes, laboratories, and stakeholders across fundamental, atomic and nuclear physics, astrophysics, environmental science, biology, life sciences, detector development, solid-state physics, imaging, medical physics, raytracing simulations, agrifood, material science, nuclear forensic, radiation security, cultural heritage, optics, photon sources, and electronics.

- Survey the globally most widely used photon detection techniques, identifying their strengths, weaknesses, and potentials for improvement based on interest range, energy, resolutions, and counting capabilities.
- Create a collection of private enterprises, including manufacturers and other stakeholders, to facilitate collaboration on market applications, new tools for data acquisition and analysis, and user-friendly software development.
- Promote knowledge exchange among Monte Carlo simulation communities, integrating ray-tracing software with other physics simulations (e.g., GEANT4, FLUKA).
- Support the comparison and exchange of advanced data fitting and simulation techniques, focusing on novel Machine Learning algorithms. This includes targeted training and expert exchanges to standardize effective data analysis techniques.
- Compare and exchange the latest techniques in development of readout electronics for photon detectors identifying weak points and possible spaces for improvement.
- Identify Grand Societal Challenges that ENRICH can address through technological improvements, outlining current obstacles and linking challenges to the appropriate Working Groups.
- Create a web platform for research funding publication and application, facilitating joint projects and enhancing funding opportunities.
- Develop a web platform for optimizing beamline usage, allowing users to find compatible research teams and secure shared experimental time slots.
- Coordinate a pool of global experts to guide younger scientists and researchers and improve their knowledge, access to infrastructures, and collaboration with experienced groups and companies.

#### Capacity Building

- Serve as a transnational platform to gather and disseminate state-of-the-art knowledge in radiation detection fields, including fundamental, atomic and nuclear physics, astrophysics, environmental science, biology, life sciences, detector development, solid-state physics, imaging, medical physics, raytracing simulations, agrifood, material science, nuclear forensic, radiation security, cultural heritage, optics, photon sources, and electronics.
- Engage COST Inclusiveness Target Countries (ITC), new research groups, and teams from countries with less capacity, ensuring gender balance and supporting young researchers to provide equal access to scientific and technological knowledge.
- Strengthen collaborations among leading international scientists in radiation-related fields to develop a joint research agenda.
- Enhance collaboration between research teams and stakeholders to promote mass production of technological outcomes (e.g., detectors, electronic boards, photon sources), reducing costs and increasing accessibility.
- Develop a joint research agenda for the identified four main technological strands, focusing on scientifically and socio-economically relevant photon detection applications.
- Establish a permanent international pool of experts to assist researchers, particularly younger ones, in preparing and submitting research projects for various funding calls (e.g., ERC grants, Marie Curie grants), ensuring financial support for the Action's objectives.

## TECHNICAL ANNEX

### 1. S&T EXCELLENCE

#### 1.1. SOUNDNESS OF THE CHALLENGE

##### 1.1.1. DESCRIPTION OF THE STATE OF THE ART

The ENRICH (European Network for radiation-detection based Research and Innovation addressing increasing societal CHallenges) COST Action *aims to build a European network of researchers and stakeholders working on radiation-detection based experiments and applications.*

Photon detection research is expanding fast, and further progress demands better resolution (energy, spatial, and time), efficient optics, advanced simulation tools, and multimodality data. Large-scale infrastructures like X-ray telescopes, free-electron lasers, and 4th generation light sources are being built or upgraded. On a smaller scale, brighter photon sources are emerging, using innovative technologies like liquid anodes or new concepts of particle accelerators and colliders, e.g., driven by high-power lasers. This drives innovation in detection, optics, and data analysis technologies.

Despite the progress, *Europe faces uncoordinated activities within siloed sub-communities* (e.g., imaging, nuclear physics, synchrotron users), leading to minimal and non-efficient cross-interaction. This hampers knowledge sharing and slows the transfer of technology from large-scale infrastructures to smaller labs and industries. Detector markets are similarly segmented, divided by domain, region, or academic/industrial focus.

*Overcoming this fragmentation could yield competitive, high-quality solutions.*

The scientific and technological advancements produced by the ENRICH COST Action will allow targeting key societal Grand Challenges Europe is facing. These challenges raise crucial questions, for example:

- Industrial and energy autonomy: *“can we increase battery autonomy, environmental compliance, and lifespan?”*; *“can we make buildings and machines energetically self-sustainable?”*.
- Food security: *“is the food I eat safe? High quality? Can I check it myself?”*.
- Healthcare and prevention: *“Can we shorten diagnostic times?”*, *“Can we detect smaller tumors for better treatment?”*.
- Risk of pandemics: *“Can we reduce research and testing time for new medicines and vaccines?”*.
- Nuclear forensic and monitoring: *“Can we fast and efficiently detect dirty bombs or radiation hazards at the national borders or in critical places like airports?”*

These challenges can indeed only be tackled through multidisciplinary scientific investigations, where analytical science significantly impacts every study. Analytical science covers imaging, structural, and chemical analysis. In this context, X-rays (0.1 keV to 100s keV) and Gamma-rays (up to 1 MeV) are invaluable for non-destructive investigation of matter. Their penetrating power and ability to excite atomic core levels allow for detecting signatures of their environment, with direct impacts on crucial societal aspects like tumor detection (X-ray imaging), swift formulation of vaccines (crystallography), food quality, security, and counterfeiting prevention (XRF for elemental analysis and chemical speciation), and many others. Overcoming the present technological limits in photon detection and related applications is the only possible way towards a successful tackling of Grand Societal Challenges.

Within the ENRICH Cost Action, four main investigation strands have been identified and reported below including, for each one, a non-exhaustive list of critical aspects to face.

**Photon Sources and Optics:** Synchrotrons and X-ray free-electron lasers (XFELs) are the gold standard for X-ray measurements, but they are expensive, scarce, and access is highly competitive. Tabletop photon sources are a common alternative, but their lower X-ray beam quality (e.g., reduced flux or tunability) can hinder measurements, especially in low-yield applications like liquid sample investigations. For example, high-resolution measurements of the  $K\beta$  transition line of metallic compounds, which provide insights into the oxidation and spin states of metals, require energy resolution at the level of a few eV. This can only be achieved using expensive Transition Edge Sensors (TES) or Metallic Magnetic Calorimeters (MMC) devices or crystal spectrometers. Applications such as food quality control demand such measurements on liquid samples, where self-absorption dominates, leading to low signal rates. This necessitates either high-brilliance facilities, extended measurement times in local labs or a simpler treatment of the samples to significantly improve the sensitivity down to 1ppb for conventional table-top EDXRF systems.

A major obstacle in photon detection is the current X-ray optics, which are costly and have low efficiency, especially at small spot sizes/high resolutions and higher energies. This limits the translation of

techniques from large-scale facilities to smaller labs or to crucial locations such as hospitals, industries,  
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and universities.

**Photon detection:** One major challenge in photon detection is the lack of a single device that offers together high energy ( $<0.1\%$ ), spatial (a few microns) and time resolution (a few ns or below), and high detection efficiency. Current technology offers energy resolution of a few %, using silicon detectors (SDD) for X-rays below 20 keV, CdZnTe detectors up to a few hundred keV, and HPGe detectors for higher energies. Bragg/Laue spectroscopy achieves much better resolution, but suffers from poor geometrical acceptance, significantly lower than solid-state devices. Micro-calorimeters like TES or MMC offer much higher resolution with reasonable acceptance but require complex, expensive cryogenic systems, operating at very low temperatures (mK range), limiting their widespread use.

Cryogenic cooling is also required for other high-performance detectors, significantly raising costs and imposing constraints on experimental setups, particularly in limited spaces like accelerators.

Energy and spatial resolution tend to be inversely proportional, meaning high spatial resolution comes at the cost of energy resolution and vice versa. This impacts techniques like Wavelength Dispersive X-ray Fluorescence (WDXRF), and improving both resolutions would enhance noise reduction, and the information gathered from a single measurement.

**Electronics and Data Acquisition (DAQ):** Pulse processing techniques are inefficient at very high fluxes, where high-throughput detection systems are required. On the opposite site, rare events and low-brilliance sources face the challenge of single photon counting. To overcome both these limitations, a big opportunity for novel hardware and software solutions to develop low-noise detectors and advanced denoising techniques is brought forward by the interaction between physics and Machine Learning (ML). Achieving cutting-edge photon detection with combined energy, time, and spatial resolutions requires a collaborative effort across research teams in solid-state device production (including new materials) and analog and digital electronics development.

Low-noise analog front-end electronics are needed for sub-keV room temperature energy resolution, complemented by digital electronics for optimal pulse processing and multi-parameter analysis.

Another challenge is the large volume of data produced by high-quality, fast, and high-resolution measurements. This “data deluge” issue is time-consuming and labor-intensive to process, especially in imaging applications, where advancements in pixel size and density lead to exponential data growth, and it should be also attacked at the DAQ level, implementing new and faster boards allowing, thanks to novel ML algorithms, for a better online data quality and rejection.

**Data Analysis, Software and Ray Tracing simulations:** The widespread use of Machine Learning (ML) techniques in data analysis, image reconstruction, and beam-line control presents challenges, particularly regarding the need for standardizing data analysis methods and ensuring a broader understanding of the underlying algorithms to prevent biases. However, ML offers faster and more accurate software solutions to many experimental problems like, for example, PLS regression which is used to improve quantification of the results.

For example, experiments with optical elements like slits, mirrors, and crystals require precise alignment and optimization, which can be time-consuming, especially in low-photon-count applications. At beamline facilities, where time is limited, long optimization periods can hinder successful measurements.

Additionally, photon detection experiments heavily rely on ray-tracing simulations, which predict beam dimensions, intensity, and energy. Many experiments require input from external simulation tools, such as MC simulation codes, to provide physical parameters. For example, using a Bragg spectrometer to measure photons resulting from particle interactions in an accelerator requires a complex MC simulation to determine the input source parameters.

Addressing the outlined issues will be accelerated only by *developing a broader community that exchanges information and expertise*. This Action's multidisciplinary nature *will drive progress in addressing Grand Challenges* like energy crises, climate change, and health. By fostering open and inclusive collaboration across these fields, *ENRICH will facilitate cross-disciplinary and international research, enhancing the effectiveness and cost-efficiency of global scientific projects in line with the COST Action's vision and spirit*.

#### 1.1.2. DESCRIPTION OF THE CHALLENGE (MAIN AIM)

The ENRICH COST Action aims *to strengthen connections* among different radiation detection physics sub-communities, *creating a cross-border, integrated community* where *multidisciplinary knowledge, information exchange, and discussion of new ideas and challenges* are central. ENRICH will contribute to address the issues identified in Section 1.1.1 by promoting collaboration among existing communities to achieve significant advancements beyond the current state of the art, as follows:



**Photon Detection:** The ENRICH action aims to advance technological research in radiation detection, focusing on developing new detectors, materials and detection techniques to enhance energy, spatial, and time resolutions beyond current technologies. Emphasis will be given to achieving top performance at non-cryogenic temperatures and overcoming limitations in combining cutting-edge resolutions. ENRICH seeks to facilitate the exchange of information and expertise among various sub-communities to create next-generation devices and techniques. ENRICH seeks to facilitate the exchange of information and expertise among various sub-communities to create next-generation devices and techniques as well as motivate small and medium size enterprises (SME) for producing photon detection systems based on the new developed technologies.

**Photon Sources and Optics:** ENRICH aims to advance research on tabletop high-flux photon sources, optical elements, and motorized positioners and holders in collaboration with stakeholders and producers. Innovations in materials, design, or production methods will make optics more affordable and efficient, broadening the use of X-rays. At the same time, sharing knowledge among researchers and producers will allow for the realization of more powerful and efficient tabletop photon sources respecting the security standards.

**Data Analysis, Software, and Ray Tracing Simulations:** The ENRICH Action will integrate the raytracing/wave optics simulations community, which currently focuses on beamline applications, into the broader radiation detection field. The aim is to develop high-quality ray propagation simulation codes for general applications and assist research groups in applying these codes. ENRICH will also seek to enhance collaboration between the raytracing/wave optics community and other MC simulation frameworks like, for example, GEANT and FLUKA, promoting multidisciplinary experiments. The Action will leverage Machine Learning to standardize analysis techniques for photon detection applications, addressing challenges like handling large data volumes and improving experimental reliability. Additionally, ENRICH will aim to enhance alignment and optimization of experimental setups through encouraging collaborations towards new automatic self-optimization software using Machine Learning algorithms.

**Electronics & DAQ:** ENRICH will help develop novel hardware and software for advanced detection techniques, fostering joint efforts to improve noise suppression, efficiency at high fluxes, and resolution in energy, space, and time. Collaboration with stakeholders such as electronic board designers will be crucial, with the aim of improving the implementation of Machine Learning algorithms in the DAQ boards. ENRICH aims to develop the new generation readout electronics by reducing noise in front-end electronics and developing new pulse processing techniques. This includes creating fast, low-noise charge-sensitive preamplifiers and digital pulse processing electronics, which offer benefits such as custom filters, stability, and multi-parameter analysis. Enhanced collaboration between researchers in both detectors and electronics development will improve performance and integration.

Simultaneous advancements in these areas will address societal challenges such as, for example: battery autonomy (by having a better chemical-physical analysis of cathode and anode materials), food safety (by identifying tiny traces of dangerous metals), early disease detection (by improving spatial, time and energy resolutions for better imaging quality), faster medical diagnostics (by reducing the necessary doses and enhancing the precision tissues investigation with more performant photon detection).

Various groups of subjects will benefit from the ENRICH Action:

**Scientists, researchers, and PhD students** across fields like Physics, Biology, Medicine, Agrifood, Chemistry, Environmental Science, Computing, Engineering, Life Science, and many others. They will gain advanced knowledge in photon detection techniques and applications, enhancing their teamwork and problem-solving skills for scientific and technological challenges.

**Stakeholders**, including existing networks, academic and research institutes, healthcare providers, public and civil organizations, and producers of photon detectors, optical components, new materials for photon detection, and laboratory equipment. They will directly connect with potential end users, discuss technological solutions, and gather information on critical issues and needs to guide their research directions.

**The European and non-European populations** will benefit from technological advancements enabled by ENRICH, leading to improved techniques for food quality assessment, medical care, prevention, and faster responses to energy, climate, and health crises.

## 1.2. PROGRESS BEYOND THE STATE OF THE ART

### 1.2.1. APPROACH TO THE CHALLENGE AND PROGRESS BEYOND THE STATE OF THE ART

The ENRICH Action will build a collaborative and multidisciplinary network of international experts in radiation detection, aiming to create a broad, interdisciplinary community where *many other scientists will benefit from individual research expertise*. The goal is to *advance photon detection science*, impacting fields such as particle and nuclear physics, astrophysics, foundational physics, environmental science, agrifood, medicine, chemistry, cultural heritage and many others.

This *expanded community* will disseminate and gather scientific knowledge, discuss recent results and needed improvements, collaborate on experiment design, train young researchers, and *attract funding through application to national and international calls*.

ENRICH will partner with leading industries to support experimental work and develop new tools, software, and components based on research outcomes.

Technological Efforts (TE) ENRICH aims to pursue, to overcome the present state of art in radiation detection applications, by fostering discussion and the interchange of know-how and new ideas among the various communities, include:

**TE1:** Reduce the gap between energy resolution, efficiency, and cost for radiation detectors by exploring new materials for photon detection, configurations, and readout techniques.

**TE2:** Enhance Research & Development (R&D) for photon detectors with high energy, space and time resolution at room or non-cryogenic temperatures.

**TE3:** Develop more efficient and cost-effective optics through new manufacturing processes and materials.

**TE4:** Improve the brilliance and power of lab-based photon sources like X-ray tubes.

**TE5:** Integrate advanced raytracing and wave optics simulations with MC simulation software to enhance multidisciplinary experiments.

**TE6:** Bring innovations in data acquisition and pre-processing hardware and software, including AI-powered solutions.

**TE7:** Standardize new Machine Learning data analysis techniques for broader understanding and use.

**TE8:** Develop Machine Learning-based self-alignment tools for optical components adaptable to various applications and measurements.

The above-mentioned TE will be crucial to the development of new tools and instrumentations for several societal applications, like improved medical diagnostic machines.

*The ENRICH vision aligns with the COST Action's emphasis on knowledge sharing. COST support will be crucial for networking activities, including meetings, training schools, conferences, workshops, and Short-Term Scientific Missions (STSMs), with a priority on training young researchers.*

### 1.2.2. OBJECTIVES

#### 1.2.2.1. Research Coordination Objectives

To address the ENRICH Action main challenges and achieve its technological goals, the following Research and Coordination Objectives have been identified:

**RCO1:** Build an overtime increasing broad and multidisciplinary community that includes experts, institutes, laboratories, and stakeholders across fundamental, atomic and nuclear physics, astrophysics, environmental science, biology, life sciences, detector development, solid-state physics, imaging, medical physics, raytracing simulations, agrifood, material science, nuclear forensic, radiation security, cultural heritage, optics, photon sources, and electronics. This includes cataloguing existing but unconnected sub-communities.

**RCO2:** Survey the globally most widely used photon detection techniques, identifying their strengths, weaknesses, and potentials for improvement based on interest range, energy, resolutions, and counting capabilities.

**RCO3:** Create a collection of private enterprises, including manufacturers and other stakeholders, to facilitate collaboration on market applications, new tools for data acquisition and analysis, and user-friendly software development.

**RCO4:** Promote knowledge exchange among Monte Carlo simulation communities, integrating raytracing software with other physics simulations (e.g., GEANT4, FLUKA).

**RCO5:** Support the comparison and exchange of advanced data fitting and simulation techniques, focusing on novel Machine Learning algorithms. This includes targeted training and expert exchanges to



standardize effective data analysis techniques.

**RCO6:** Compare and exchange the latest techniques in development of readout electronics for photon detectors identifying weak points and possible spaces for improvement.

**RCO7:** Identify Grand Societal Challenges that ENRICH can address through technological improvements, outlining current obstacles and linking challenges to the appropriate Working Groups.

**RCO8:** Create a web platform for research funding publication and application, facilitating joint projects and enhancing funding opportunities.

**RCO9:** Develop a web platform for optimizing beamline usage, allowing users to find compatible research teams and secure shared experimental time slots.

**RCO10:** Coordinate a pool of global experts to guide younger scientists and researchers and improve their knowledge, access to infrastructures, and collaboration with experienced groups and companies.

#### 1.2.2.2. Capacity-building Objectives

In parallel with the Research Coordination Objectives, the following Capacity-building Objectives are essential for addressing the ENRICH Action's main challenges and achieving its technological goals:

**CBO1:** Serve as a transnational stakeholder platform to gather and disseminate state-of-the-art knowledge in radiation detection fields, including fundamental, atomic and nuclear physics, astrophysics, environmental science, biology, life sciences, detector development, solid-state physics, imaging, medical physics, raytracing simulations, agrifood, material science, nuclear forensic, radiation security, cultural heritage, optics, photon sources, and electronics. This CB aims as well to gather and disseminate information on infrastructures, research groups, and manufacturers.

**CBO2:** Engage COST Inclusiveness Target Countries (ITC), new research groups, and teams from countries with less capacity, ensuring gender balance and supporting young researchers to provide equal access to scientific and technological knowledge.

**CBO3:** Strengthen collaborations among leading international scientists in radiation-related fields to develop a joint research agenda.

**CBO4:** Enhance collaboration between research teams and stakeholders to promote mass production of technological outcomes (e.g., detectors, electronic boards, photon sources), reducing costs and increasing accessibility.

**CBO5:** Develop a joint research agenda for the identified four main technological strands, focusing on scientifically and socio-economically relevant photon detection applications.

**CBO6:** Establish a permanent international pool of experts to assist researchers, particularly younger ones, in preparing and submitting research projects for various funding calls (e.g., ERC grants, Marie Curie grants), ensuring financial support for the Action's objectives.

*These Research Coordination and Capacity-building Objectives will adhere to the COST principles of inclusivity and excellence, encouraging young researchers' involvement and maintaining gender balance, while maximizing participation from emerging countries.*

## 2. NETWORKING EXCELLENCE

### 2.1. ADDED VALUE OF NETWORKING IN S&T EXCELLENCE

#### 2.1.1. ADDED VALUE IN RELATION TO EXISTING EFFORTS AT EUROPEAN AND/OR INTERNATIONAL LEVEL

There are numerous European and international projects focused on photon detection, each targeting specific sectors. Examples include initiatives for radiation protection (EURATOM, EURADOS, MELODI), medical research (Euro-Biolmaging, MEDIRAD), environmental monitoring (NORMAN), nuclear fusion (EUROfusion), and training (ENEN). These projects, however, aim to advance scientific knowledge and develop new technologies, but each operates within its own domain.

The ENRICH Action will *complement these efforts* by enhancing knowledge sharing, fostering interdisciplinary collaboration, and driving innovation through cross-community exchanges. Key to this effort will be *engaging active participants from existing networks to join ENRICH*, which will facilitate a comprehensive census of photon detection research teams, identify suitable intervention areas, and quickly advance ENRICH objectives.

*Existing networks will benefit from improved collaboration and knowledge sharing through ENRICH as well.*

The ENRICH Cost Action added value can be summarized in three main pillars:

**Integration of efforts:** ENRICH will unify existing projects and networks, promoting the exchange of ideas, data, and methodologies. This will enable researchers to build on each other's work and foster interdisciplinary collaboration across fields like fundamental, atomic and nuclear physics, astrophysics, environmental science, biology, life sciences, detector development, solid-state physics, imaging, medical physics, raytracing simulations, agrifood, material science, nuclear forensic, radiation security, cultural heritage, optics, photon sources, and electronics.

**Resource Sharing and Finding:** By creating a network, ENRICH will optimize the use of sophisticated equipment and facilities, such as synchrotrons and specialized software. Coordinated scheduling and shared resources will make advanced technology more accessible and ensure broad benefits for researchers. To ensure the success of the technological and scientific advancements proposed by ENRICH, dedicated tasks will be devoted to helping the participating researchers in submitting proposals for national and international funding applications.

**Accelerated Innovation:** ENRICH will speed up the development and application of new technologies through fostering joint research projects and collective expertise. This will lead to rapid advancements in photon radiation techniques and applications, including the realization of new and more performant tools for societal applications, like devices for medical inspections or food control.

In summary, ENRICH will enhance existing photon detection research efforts by fostering synergy, optimizing resources, and accelerating innovation. *This cohesive network will improve the impact and efficiency of research, benefiting healthcare, environmental monitoring, and industrial applications.*

## 2.2. ADDED VALUE OF NETWORKING IN IMPACT

### 2.2.1. SECURING THE CRITICAL MASS, EXPERTISE AND GEOGRAPHICAL BALANCE WITHIN THE COST MEMBERS AND BEYOND

The success of the ENRICH COST Action depends on assembling a diverse and balanced network of experts and institutions across Europe, with a strategy that involves three key elements:

**Engaging Leading Experts:** ENRICH will identify and invite top researchers in photon radiation from various fields, including fundamental, atomic and nuclear physics, astrophysics, environmental science, biology, life sciences, detector development, solid-state physics, imaging, medical physics, raytracing simulations, agrifood, material science, nuclear forensic, radiation security, cultural heritage, optics, photon sources, and electronics. Existing networks will be leveraged to recruit both renowned and emerging talents, while mentoring programs and trainings will be organized to provide guidance and facilitate connections.

**Promoting Inclusiveness:** ENRICH prioritizes inclusiveness by involving researchers from more than 20 countries with an almost equal gender balance already at the initial stage. Moreover, special attention will be given to supporting young researchers and women, as well as to those with caregiving responsibilities through dedicated activities and childcare facilities. Participation in events for researchers with caregiving duties will be encouraged by establishing a series of dedicated activities like, for example, offering in-situ childcare facilities.

**Ensuring Geographical Balance:** To ensure all participating countries are represented, ENRICH will monitor and adjust activities for fair distribution. Transparent processes and clear criteria will be established to ensure equal access to funding and resources, as well as to leading roles in the Action Management and in the Working Groups.

*By engaging top experts, promoting inclusiveness, and ensuring geographical balance, ENRICH will create a dynamic network that drives innovation and addresses key challenges in photon radiation, benefiting Europe and beyond.*

### 2.2.2. INVOLVEMENT OF STAKEHOLDERS

The involvement of stakeholders is critical to the success and impact of the ENRICH COST Action on photon detection and related applications. Engaging a diverse range of stakeholders ensures relevance, addresses real-world challenges, and facilitates the translation of research into practical applications.

Key stakeholder groups include:

**Academic and Research Institutions:** Universities, research centers, and laboratories in photon radiation, nuclear physics, medical imaging, material science, cultural heritage and related fields.

**Industry Partners:** Companies developing and applying radiation technologies, including medical device manufacturers and environmental monitoring firms.

**Healthcare Providers:** Hospitals, clinics, and practitioners using radiation technologies for diagnostics

and treatment.

**Regulatory Bodies and Policy Makers:** Agencies regulating radiation safety, medical devices, and environmental protection.

**Public and Civil Society Organizations:** Patient advocacy groups, environmental NGOs, and organizations representing public interests.

Stakeholder engagement will focus on three main pillars:

**Regular Communication and Consultation:** ENRICH will maintain open communication through regular Stakeholder Meetings, Workshops, and Webinars to update stakeholders, gather feedback, and discuss future directions. Stakeholders will be also invited to participate to the Working Groups meetings and activities.

**Collaborative Research and Development Projects:** ENRICH will promote discussions on R&D projects involving stakeholders to ensure relevance and practical application. Joint projects with industry stakeholders and collaborations with healthcare providers will focus on co-developing technologies and refining tools and devices for all practical applications like, for example, medical imaging and treatment technologies.

**Dissemination and Exploitation of Results:** A comprehensive plan will ensure widespread and effective dissemination of research findings. This includes supporting and patronizing workshops and conferences, organizing targeted training sessions, and developing policy briefs to inform regulatory bodies and policy makers.

By involving a diverse range of stakeholders, *ENRICH will ensure that its research is relevant and impactful, facilitating the rapid translation of discoveries into practical applications and maximizing societal and economic benefits.*

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### 3. IMPACT

#### 3.1. IMPACT TO SCIENCE, SOCIETY AND COMPETITIVENESS, AND POTENTIAL FOR INNOVATION/BREAKTHROUGHS

##### 3.1.1. SCIENTIFIC, TECHNOLOGICAL, AND/OR SOCIOECONOMIC IMPACTS (INCLUDING POTENTIAL INNOVATIONS AND/OR BREAKTHROUGHS)

The ENRICH COST Action is set to make a significant impact across scientific, technological, and socioeconomic domains. By fostering interdisciplinary collaboration and knowledge sharing, it will drive innovation and breakthroughs addressing critical challenges. In Fig. 1, a schematic of the main anticipated impacts and of the measures to maximize impact of the ENRICH Action are highlighted.

More in detail, the anticipated impacts of the ENRICH Action include:

**Scientific Impacts:** The ENRICH Action will advance fundamental knowledge of photon radiation phenomena, including interactions with matter at atomic, nuclear and molecular levels. Integrating expertise from diverse scientific fields will yield novel insights into photon applications in fundamental, atomic and nuclear physics, astrophysics, environmental science, biology, life sciences, detector development, solid-state physics, imaging, medical physics, raytracing simulations, agrifood, material science, nuclear forensic, radiation security, cultural heritage, optics, photon sources, and electronics. The promotion of standardized experimental techniques will enhance the reliability and reproducibility of results, and collaborative research will foster the development of new analytical tools and technologies.

**Technological Impacts:** ENRICH will significantly influence the field of radiation detectors by developing advanced detectors with improved capabilities and performances at room temperature. This will enable smaller, cost-effective experimental setups, devices and tools. Innovations in low-noise electronics and advanced pulse processing will facilitate room temperature measurements under high flux conditions. The development of cutting-edge detection technologies will enable, for example, early detection of diseases and reduce exposure times in medical exams. Additionally, industrial applications will benefit from enhanced radiation detection technologies, optimizing processes like non-destructive testing and material analysis crucial, for example, in many cultural heritage or food control applications.

**Socioeconomic Impacts:** Economic growth and job creation will stem from the development of new photon radiation technologies, creating market opportunities in high-tech sectors. New analytical techniques and Machine Learning software will aid in identifying counterfeit products, preventing economic losses. The commercialization of research findings will generate high-skilled jobs in research, development, and manufacturing, while SME will be motivated in producing photon detection systems based on the new developed technologies. For example, advances in medical imaging and radiation therapy will improve diagnosis and patient care, while enhanced environmental monitoring will safeguard

communities from radiation hazards. Training programs and workshops will elevate skills and expertise, especially among early-career scientists and those from ITCs, raising the overall quality of research and education.

**Potential Innovations and Breakthroughs:** ENRICH will lead to the development of novel machines, tools and devices for several societal applications, like for example new diagnostic and therapeutic tools for early disease detection and effective treatments. It will also drive the discovery of new materials with unique properties, enhancing radiation detection devices and providing new energy solutions, such as longer-lasting batteries. The expected increased analytical capabilities of benchtop XRF systems or handheld units will allow non-invasive analysis of samples in perfect agreement with the Green Analytical Chemistry (GAC) postulates.

Overall, the ENRICH COST Action will drive significant advancements in photon radiation research, *positioning Europe as a global leader* and benefiting society through *improved health, safety, and economic growth*.

## 3.2. MEASURES TO MAXIMISE IMPACT

### 3.2.1. KNOWLEDGE CREATION, TRANSFER OF KNOWLEDGE AND CAREER DEVELOPMENT

The ENRICH COST Action aims to foster knowledge creation, facilitate knowledge transfer, and promote career development among researchers. By building a collaborative and inclusive network, the Action will enhance scientific expertise, disseminate cutting-edge research findings, and support member professional growth.

The strategies to achieve these goals are outlined below.

**Knowledge Creation:** The Action will support collaborative research projects across diverse disciplines and institutions. These projects will address key scientific and technological challenges in photon radiation research, leading to innovative solutions. By integrating perspectives from detectors and electronics development, astrophysics, atomic, nuclear and hadronic physics, medical imaging, material science, and other fields, ENRICH will promote interdisciplinary research that advances the state of the art. Access to research infrastructure will be optimized; the ENRICH network will facilitate access to shared facilities (like synchrotrons), radiation detectors, and computational tools, enabling high-quality experiments and analyses. Centralized data repositories will enhance the reproducibility and impact of scientific findings through efficient data sharing.

**Transfer of Knowledge:** The Action will prioritize disseminating research results through high-impact and open access publications, conference presentations, and journal special issues to reach the broader scientific community. Regular workshops, seminars, and webinars will share research findings, best practices, and technological innovations.

Knowledge Exchange Platforms, such as websites and online collections, will facilitate information sharing. Collaborative tools like virtual workspaces will enable real-time interaction among network members. Outreach to stakeholders will be ensured through joint projects, technology transfer agreements, and innovation hubs, translating research into practical applications. Developing policy briefs and engaging with regulatory bodies will help shape policies that support the safe use of radiation detection technologies.

**Career Development:** Training will be central to ENRICH. The Action will offer workshops, summer schools, and short courses to enhance researchers' skills, particularly those from Inclusiveness Target Countries (ITCs). Mentorship schemes will pair early-career researchers with experienced mentors for guidance and development.

Mobility and Exchange Opportunities will be crucial. Short-term research visits and secondments will provide experience in different laboratories and access to specialized equipment. Exchange programs will foster cross-institutional collaboration and cultural exchange. Career Advancement Support will include workshops on grant writing, project management, and career planning, alongside recognition and award programs to motivate researchers.

**Attract fundings:** Ensuring the necessary financial resources for the various R&D activities will play a central role in ENRICH. The Action will establish a board of experts to help researchers in successfully writing scientific proposals by offering workshops, courses and mentoring sessions. This will allow to attract fundings for the R&D activities and will also serve as career boosting opportunity for young researchers.

By building a dynamic and inclusive network, the Action will advance scientific expertise, promote research application, and support member growth, *ensuring Europe remains at the forefront of innovation in photon radiation research*.



### 3.2.2. PLAN FOR DISSEMINATION AND/OR EXPLOITATION AND DIALOGUE WITH THE GENERAL PUBLIC OR POLICY

The ENRICH COST Action is dedicated to ensuring that its findings and innovations are widely disseminated to the scientific community, industry stakeholders, policymakers, and the general public. To maximize impact and fully realize the benefits of the research, effective dissemination and exploitation strategies, combined with active dialogue, are crucial.

The following plan outlines the approach to achieving these objectives:

**Dissemination Strategy:** A key component of the strategy will focus on the Scientific Community. High-impact publications in peer-reviewed open-access journals will be a primary method for sharing research findings. Special issues and thematic collections will highlight the Action's contributions to the field. Organizing and participating in international conferences, symposia, and workshops will facilitate idea exchange and showcase the Action's research outcomes. Making research outputs, including datasets and software, available in open-access repositories will enhance visibility and accessibility.

Another significant aspect will address Industry and Stakeholders. Workshops and webinars tailored for industry partners will focus on the practical applications of research findings, promoting technology transfer and commercialization. Joint research and development projects with industry partners will support the direct application of new technologies and innovations in commercial settings. Establishing innovation hubs or incubators will aid in developing and scaling new technologies emerging from the Action's research. Establishing advisory panels with representation from regulatory bodies will provide ongoing dialogue and collaboration on policy-related issues.

**Dialogue with the General Public and Policymakers:** Special emphasis will be placed on engaging the General Public. Organizing public lectures, science cafés, open days, and community events will engage the public in discussions about photon radiation, its applications, and its societal impact. Through social media, press releases, and educational materials for schools, such as brochures, videos, and online resources, the ENRICH COST Action will provide accessible information about the research and its benefits. Leveraging traditional and social media platforms will help disseminate research findings and raise public awareness about the Action's activities and achievements.

Public Engagement Activities will include initiatives such as interactive exhibitions at science museums and public events to provide hands-on learning experiences related to photon radiation. ENRICH will try to involve the general public in research activities, fostering a deeper understanding and appreciation of scientific inquiry. For example, extensive participation in initiatives like the Research's Night across Europe will be encouraged.

Educational Outreach will be set up with school programs and materials to educate students about photon radiation detection and its applications, inspiring the next generation of scientists and informed citizens. Offering online courses and webinars on relevant topics will make educational content accessible to a broad audience, including students, educators, and the general public.

By engaging with the scientific community, industry stakeholders, policymakers, and the general public, the Action will *maximize its contributions to knowledge, technology, and society*. This strategic approach will enhance the visibility, accessibility, and application of research outcomes, fostering a collaborative and informed environment that supports the *advancement of research on photon detection and its beneficial applications*.

## 4. IMPLEMENTATION

### 4.1. COHERENCE AND EFFECTIVENESS OF THE WORK PLAN

#### 4.1.1. DESCRIPTION OF WORKING GROUPS, TASKS AND ACTIVITIES

The implementation of the ENRICH Action will be organized into seven Working Groups (WGs), collectively addressing the Research Coordination Objectives (RCO), Capacity-Building Objectives (CBO), and Technological Efforts (TE) outlined in the proposal. The interconnections among the WGs will be structured to ensure a synergistic approach, enabling the effective achievement of the Action's goals (see Fig. 2). For each WG, the specific tasks (T) are listed below with the involved RCO, CBO and TE indicated in brackets.

**WG1: Community Coordination:** this WG focuses on all the necessary activities to improve the quality of the Action and to increase the number of participants.

T1.1: Compilation of all the research teams and subcommunities initially involved in the ENRICH Action (RCO1)

T1.2: Census and compilation of existing research teams and subcommunities potentially interested in joining the ENRICH Action to promote its goals and enlarge the community (RCO1, CBO1, CBO2).

T1.3: Promote and maintain links among the various WGs by organizing regular meetings among WG leaders and disseminating the advancement of the ENRICH Action to the participants through annual reports, internal notes, websites and similar initiatives (RCO1, CBO3). A set of mailing lists and newsletters will be also created for the various WG, the management, the general public, etc.

**WG2: Technological Advancement:** this WG focuses on all the necessary activities to allow for technological advancements impacting on the various applications.

T2.1: Conduct a comprehensive survey of existing photon detection techniques, focusing on energy, position, and timing resolution capabilities (RCO2).

T2.2: Stimulate the investigation in new materials and detector configurations to enhance photon detection performances (energy, time and spatial resolution) and to develop a new generation of optical components while reducing costs (TE1, TE2, TE3).

T2.3: Connect research teams and producers to allow for a more massive production of the new technological outcomes produced by the ENRICH Action, aiming at a sizeable reduction of the production costs (RCO3, CBO4).

T2.4: Create a stronger collaboration between research teams and producers to identify the present weak points in table-top photon sources production aiming at developing and testing new photon sources improving table-top beams brilliance of, at least, one order of magnitude (TE4).

**WG3: Software Advancement:** this WG focuses on all the necessary activities involving the development of new software and standards for the community.

T3.1: Foster collaboration among Monte Carlo simulation communities to integrate ray-tracing software with other simulation tools and validate integrated simulation models for various photon detection applications (RCO4, TE5).

T3.2: Collaborate with software developers to promote the realization of open-source packages with a multidisciplinary approach, allowing for easy modifications and customizations by the research teams to be adapted to many possible applications (RCO5, TE6).

T3.3: Realize a compilation of the various data analysis techniques used by the various research teams of the Action, aiming at a more generalized standardization and wider comprehension of the experimental outcomes in different fields and applications (TE7, TE8).

T3.4 Enhance AI-powered solutions for data processing and analysis to implement next-generation data processing systems (TE8).

**WG4: Electronics and DAQ:** this WG focuses on all the necessary activities to bring readout electronics for photon detection beyond the present state of art, aiming at improved performance (energy, space and time resolutions) and reduced costs and dimensions.

T4.1: Conduct an evaluation of current readout electronics and DAQ systems, identifying strengths and weaknesses, benchmarking existing technologies against industry standards (RCO6).

T4.2: Boost the emergent technology of AI and Machine Learning based electronic boards aiming at a wider and easier implementation of these tools in many multidisciplinary applications (TE6)

T4.3: Stimulate collaboration among researchers and manufacturers in electronics to integrate novel readout solutions into photon detectors at low costs (RCO3, CBO4).

**WG5: Multidisciplinary and Societal Applications:** this WG focuses on all the activities related to multidisciplinary applications, in particular with those related to the Societal Challenges addressed by this Action.

T5.1: Compile existing photon detection techniques tailored for specific applications, identifying weak points and potential improvements for each application area (RCO2, RCO7).

T5.2: Assign each of the applications identified in T5.1 to the various WGs, with a particular focus on those addressing the Grand Societal Challenges, aiming at the realization of effective steps towards their solutions (RCO7).

T5.3: Facilitate knowledge exchange between different fields to foster interdisciplinary applications by organizing joint workshops and conferences (RCO1, RCO3, RCO5, RCO7, CBO3, CBO5, TE5).

**WG6: Integration and Interchange:** this WG focuses on all the necessary activities aimed at making all the ENRICH Action's initiatives (meetings, workshop, schools, etc.) accessible to all the researchers of the community and at attracting funds. The activities of this WG will last until the end of the COST Action and will be constantly updated to ensure the success of ENRICH.

T6.1: Implement support measures for researchers with care duties during the various events, such as offering babysitting services for example, and promoting flexible working arrangements to support researchers with diverse needs (CBO2, CBO6).

T6.2: Providing support to all the research teams participating in the Action in attracting fundings and



obtain financial supports for their activities (RCO8, CBO2)

T6.3 Coordinate access to shared experimental beamtimes and measurement campaigns (RCO9, CBO3, CBO5).

T6.4 Implement on the ENRICH Website the following Integration and Interchange Forums:

- Technology Collaboration Forum: An online platform to connect research teams and producers (demand and supply) for enhancing mass production and cost reduction.
- Open-Source Software Repository: A repository of the open-source, multidisciplinary software packages developed within the ENRICH Action.
- Electronics Collaboration Forum: An online platform to connect electronics research teams and producers (demand and supply) for enhancing mass production and cost reduction.
- Shared Experiments Forum: An online platform for coordinating shared experimental beamtimes and measurement campaigns.
- Funding Search Engine Collection: A regularly updated collection of links to the available funding search engines, divided by main categories, to collect and show the various calls for funding applications published from various funding agencies.; this platform will also serve as a publisher for funding applications to be used by the various stakeholders involved in the ENRICH Action.
- Mentoring Program Forum: An online platform for the mentoring program connecting young researchers with experienced mentors.

**WG7: Mentoring, training and dissemination:** this WG focuses on all the activities related to mentoring and training, in particular of young scientists, as well as with the dissemination of the ENRICH's outcomes both to the specialistic and to the general public.

T7.1: Establish a mentoring program to connect young researchers with experienced mentors providing regular mentoring sessions and career advice (RCO10, CBO7).

T7.2: Conduct training sessions on novel data analysis algorithms and software tools (TE7, TE8).

T7.3: Realization of the dissemination plan underlined in the previous sections and inform possible users and stakeholders on the advancements through newsletter and workshops.

T7.4: Organize webinars on writing research proposals and grant applications (CBO7, RCO8).

T7.5: Offer training on various career paths, including academia and industry, facilitating networking opportunities with industry professionals and academic leaders (RCO10, CBO1, CBO4).

In Fig. 2, a schematic of the interconnection among the tasks of the various WGs is reported.

#### 4.1.2. DESCRIPTION OF DELIVERABLES AND TIMEFRAME

The ENRICH COST Action will produce two types of deliverables: cross-sectional and WG specific ones, with delivery time defined in four-months periods (Q).

The cross-sectional deliverables will be common to all WG, and will consist in:

*DX.1 – Annual general and technical reports of the individual WG results, realized initiatives and status of the various tasks (Q5, Q9, Q13, Q16).*

*DX.2 – Annual document about the line of interventions, planned activities, goals and expected results for each WG in the following year (Q5, Q9, Q13).*

*DX.3 - Final reports and recommendations for future work, for each WG, summarizing the findings and recommendations for future research and development resulting from the ENRICH activities (Q16).*

Then, each of the WG will have specific deliverables which are listed below, with their delivery time in brackets.

#### **WG1: Community Coordination**

*D1.1 - Coordination Meeting Records (Q2, Q6, Q10, Q14):* Records and minutes from regular coordination meetings among WG leaders.

*D1.2 - ENRICH Community Census (Q5):* A comprehensive collection of research teams and subcommunities involved in the ENRICH Action, to be regularly updated along the whole duration of the Action.

#### **WG2: Technological Advancement**

*D2.1 – Census of Technology Producers (Q5):* Census of all the producers of X-ray sources, electronic components, detectors, materials and all the possible items required by the research teams involved in ENRICH.

*D2.2 – Mass Production Regulation Document (Q7):* A document including the agreements between the

various producers identified in D2.1 and the ENRICH Research Teams, where the regulations and the procedures to enhance mass production and cost lowering are outlined.

### **WG3: Software Advancement**

*D3.1 – Open-Source Software Regulation Document (Q7):* Document where the rules and the procedures for the online publication of the new software developed within ENRICH will be established.

*D3.2 - Data Analysis Standardization Document (Q16):* A final document, based on D3.2, with a proposed strategy towards a Data Analysis Standardization among the ENRICH community.

### **WG4: Electronics and DAQ**

*D4.1 – Census of Electronics and DAQ research teams (Q5):* Census of all the research teams involved in ENRICH working on front-end and readout electronics, as well as on Data Acquisition (DAQ) hardware and software.

*D4.2 - Next-Generation DAQ Systems Report (Q14):* A final report on the performance evaluation and validation of next-generation DAQ systems and readout boards.

### **WG5: Multidisciplinary and Societal Applications**

*D5.1: - Interdisciplinary Workshops Reports and Proceedings Collection (Q4):* Realization of an online collection where to store all the Minutes, Reports and Proceedings of the interdisciplinary workshops and conferences organized to facilitate knowledge exchange.

*D5.2 - Target Application List (Q6):* Compilation of the identified multidisciplinary and societal applications, related to the Grand Challenges, which can be addressed within the various WG of the COST Action.

*D5.3 - Intervention strategies and implementation plan (Q10):* Detailed plan for integrating interdisciplinary activities, including schedules and resources, including a series of documents, addressed to the various WG, where the weak points, the technological showstoppers and a proposed intervention strategy for each of the applications identified in D5.1 are reported.

### **WG6: Integration and Interchange**

*D6.1 – Disciplinary Regulation Document (Q5):* A document providing the guidelines and the regulations for the organization of all the ENRICH events, where the initiatives to be taken for the support of researchers with care duties will be explicitly outlined.

*D6.2 – ENRICH Website and mailing lists (Q7):* Website of the ENRICH Action where the main scopes of the Action, all the produced documents and the link to the various collections and platforms realized within ENRICH will be linked.

*D6.3 – Integration and Interchange Platforms (Q9):* Inclusion in the ENRICH Website of all the Integration and Interchange Forums described in the T6.4 task.

### **WG7: Mentoring, Training, and Dissemination**

*D7.1 - Training Sessions Reports (Q4, Q8, Q12, Q16):* Reports on training sessions, both for career paths, technological and data analysis techniques, including surveys and evaluations from both teachers and students.

*D7.2 - Proposal Writing Webinars Reports (Q6, Q10, Q14):* Reports on workshops conducted for writing research proposals and grant applications.

### **Milestones:**

The ENRICH Action will also have 4 cross-functional Milestones:

*M1 - ENRICH Kick-Off Meeting (Q1):* This will be the first meeting of the ENRICH Action at which the leadership positions will be elected and Action will officially start. A plan for the first year and budget distribution will be discussed and agreed upon.

*M2 - Mid-term Review Meetings (Q2, Q6, Q10, Q14):* In these meetings, that will be organized online, the coordinators of each WG will present their WGs achievements and will serve as a verification step for the various tasks and activities.

*M3 – WG Annual Meetings (Q4, Q8, Q12, Q16):* These meetings will be done in presence and will gather active participants of the various WGs to review, plan and organize the tasks and the activities of each year. These meetings will be opened to the representatives of the various stakeholders involved in each WG.

*M4 – Final Conference (Q16):* This will be the conclusive event of the ENRICH Action, where all the results will be presented and the pathways and recommendations for future works will be discussed.

### Timeline:

The timeline of the ENRICH Action spans four years, with quarterly checkpoints for tasks and deliverables. The project begins with foundational activities, such as collections' creation and platforms' development, followed by performance evaluations, training sessions, and integration plans. Annual and mid-term reviews ensure continuous monitoring and adjustments. The timeline is structured to support dependencies, such as platforms needed early to aid subsequent tasks, and concentrate deliverables around major meetings and workshops for effective coordination and output preparation.

The timeline of the various tasks and deliverables is planned to consider the following key considerations:

**First compilations and databases:** the first quarters of the ENRICH Action will be devoted to the initial activities of the various WGs and of the community in general, like the various censuses of the technologies, research teams, stakeholders, etc.

**Dependencies:** Dependencies among the various tasks are considered; for example, online platforms and repositories will be created soon after the end of the first compilations and should be completed early to support subsequent tasks of the various WGs.

**Annual and General Workshops:** The due dates of many of the deliverables, particularly those consisting in produced documents, are intentionally concentrated in particular quarters, where either Annual or Mid-term Meetings or General Workshops will take place. In these occasions, the WG members will have the occasion to meet and to prepare such documents.

*The ENRICH timeline ensures that all tasks, milestones, and deliverables are aligned, and dependencies are managed, facilitating the effective implementation of the Action.*

#### 4.1.3. RISK ANALYSIS AND CONTINGENCY PLANS

The following is a list of possible risks. For each one, a mitigation plan and the risk level are reported.

**Resource Allocation Conflicts:** Overlapping tasks and milestones across WGs could lead to resource constraints. *Mitigation:* Implement a resource management plan to ensure adequate allocation and avoid overburdening team members. *Risk Level:* Low.

**Technical Challenges:** Unforeseen technical difficulties in photon detection and data analysis could delay progress. *Mitigation:* Establish a technical advisory board to provide expert guidance and promptly troubleshoot issues. *Risk Level:* Medium.

**Collaboration Issues:** Lack of effective communication and coordination among diverse WG members. *Mitigation:* Schedule regular inter-WG meetings and use collaboration tools to ensure smooth communication and issue resolution. *Risk Level:* Medium

**Lack of fundings:** Lack of financial resources to realize the R&D activities resulting from the networking activities of ENRICH. *Mitigation:* In addition to all the activities foreseen by ENRICH for attract fundings, a resource (tools, instrumentations, laboratories, expertise, etc) sharing plan will be developed. *Risk Level:* Medium.

**Impact of External Factors:** External factors like political instability or global pandemics could disrupt project timelines. *Mitigation:* Develop a flexible project plan that can adapt to external changes and include remote work options, as well as preparing the ground for an efficient eventual organization of online meetings and events. *Risk Level:* Low

By identifying these risks and implementing the mitigation strategies, the COST Action can proceed with greater confidence in achieving its objectives on schedule.

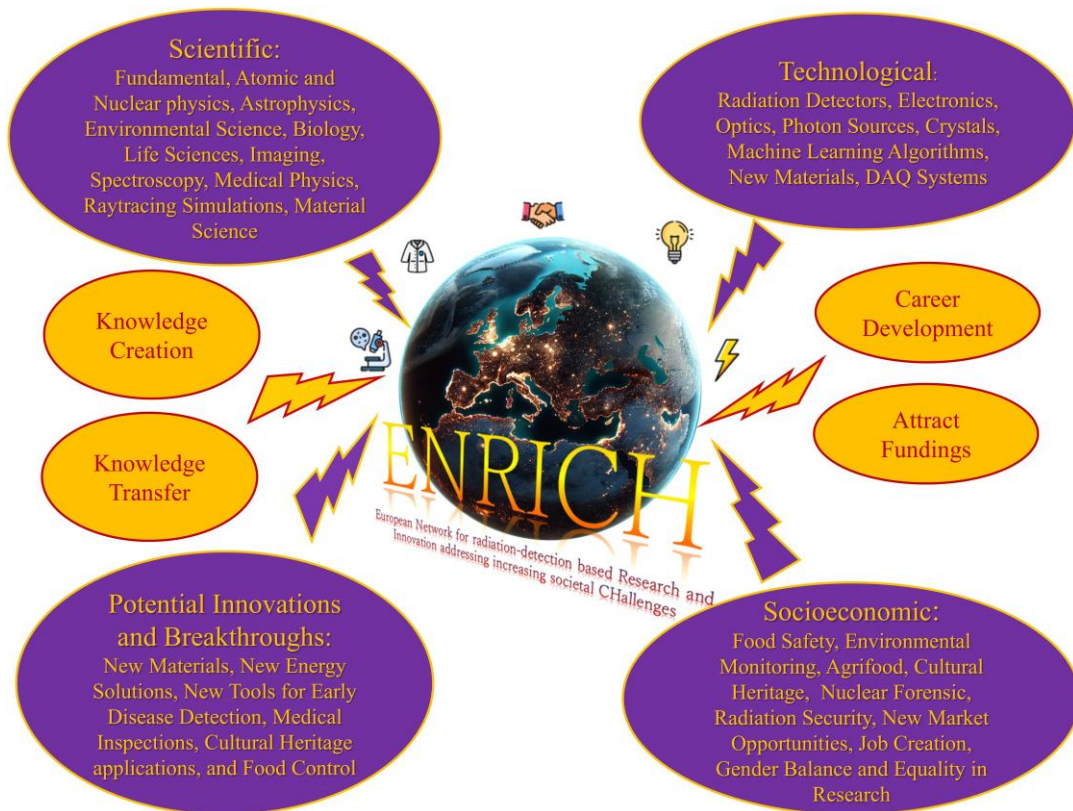


Fig. 1: Scheme of the anticipated impacts of the ENRICH COST Action

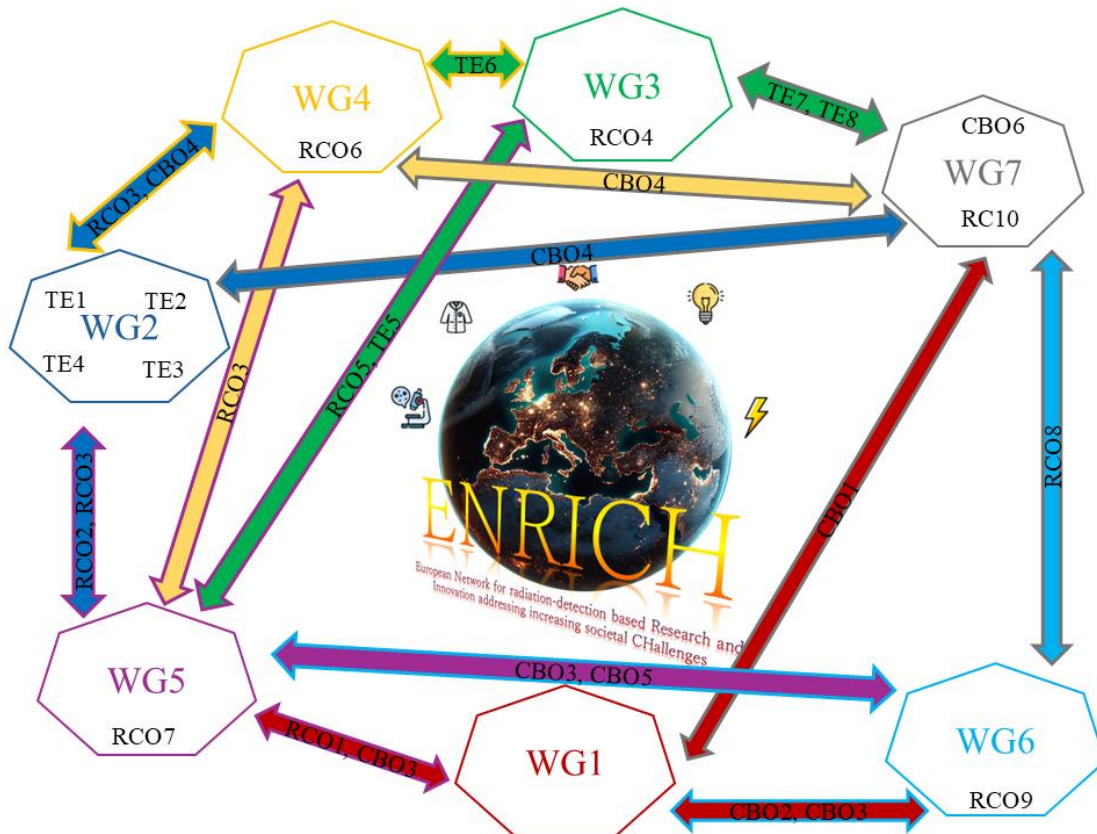


Fig. 2: Schematic of the interconnections among the different tasks of the WGs.



#### 4.1.4. GANTT DIAGRAM

In Fig. 3, the GANTT Chart of the ENRICH COST Action is reported. The timeline is divided into 16 quarters for a total of 4 years (48 months). The activities of each WG are identified by a unique color, while in the lower part the various deliverables and milestones are indicated.

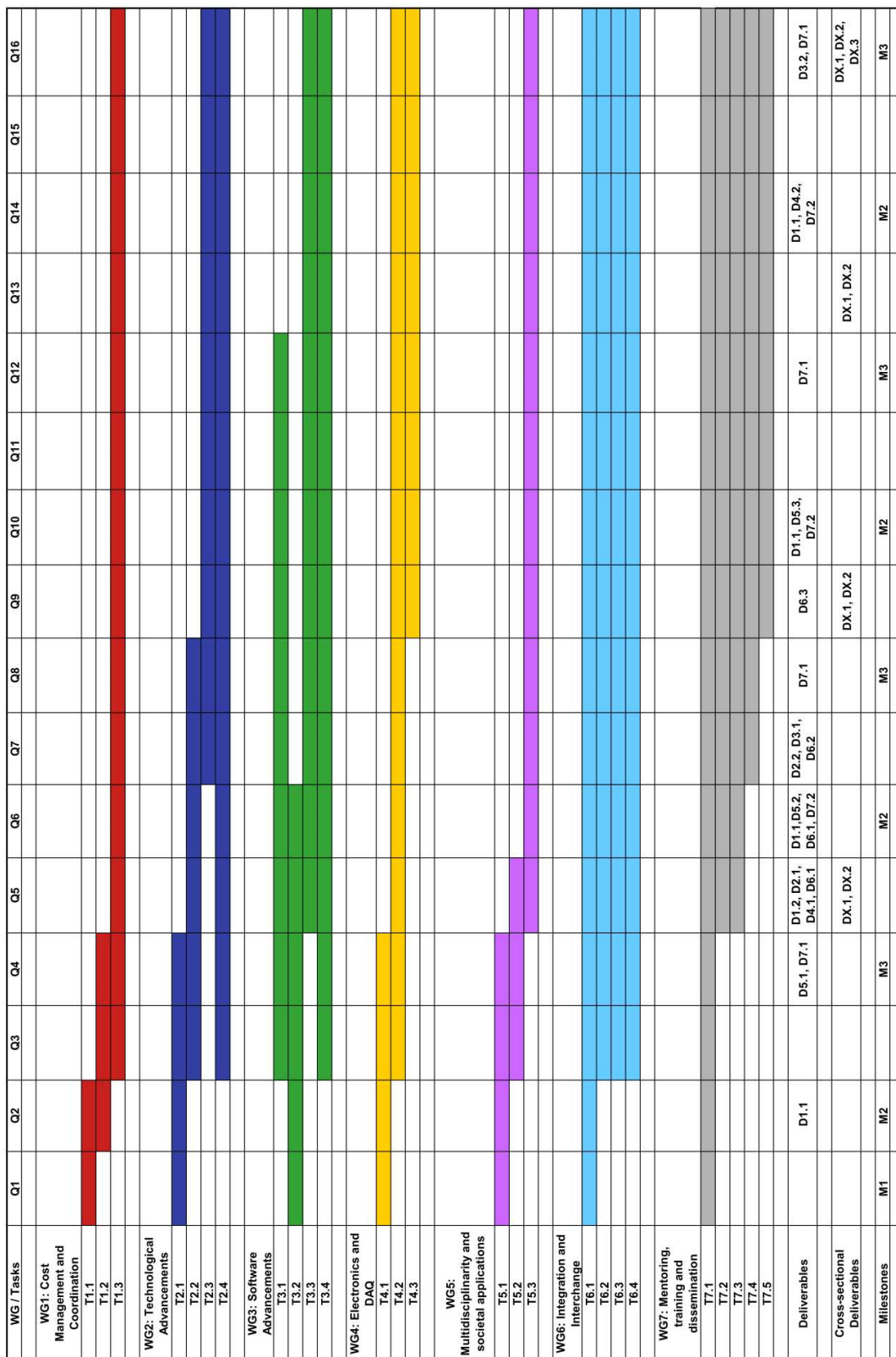


Fig. 3: GANTT Chart of the ENRICH COST Action



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